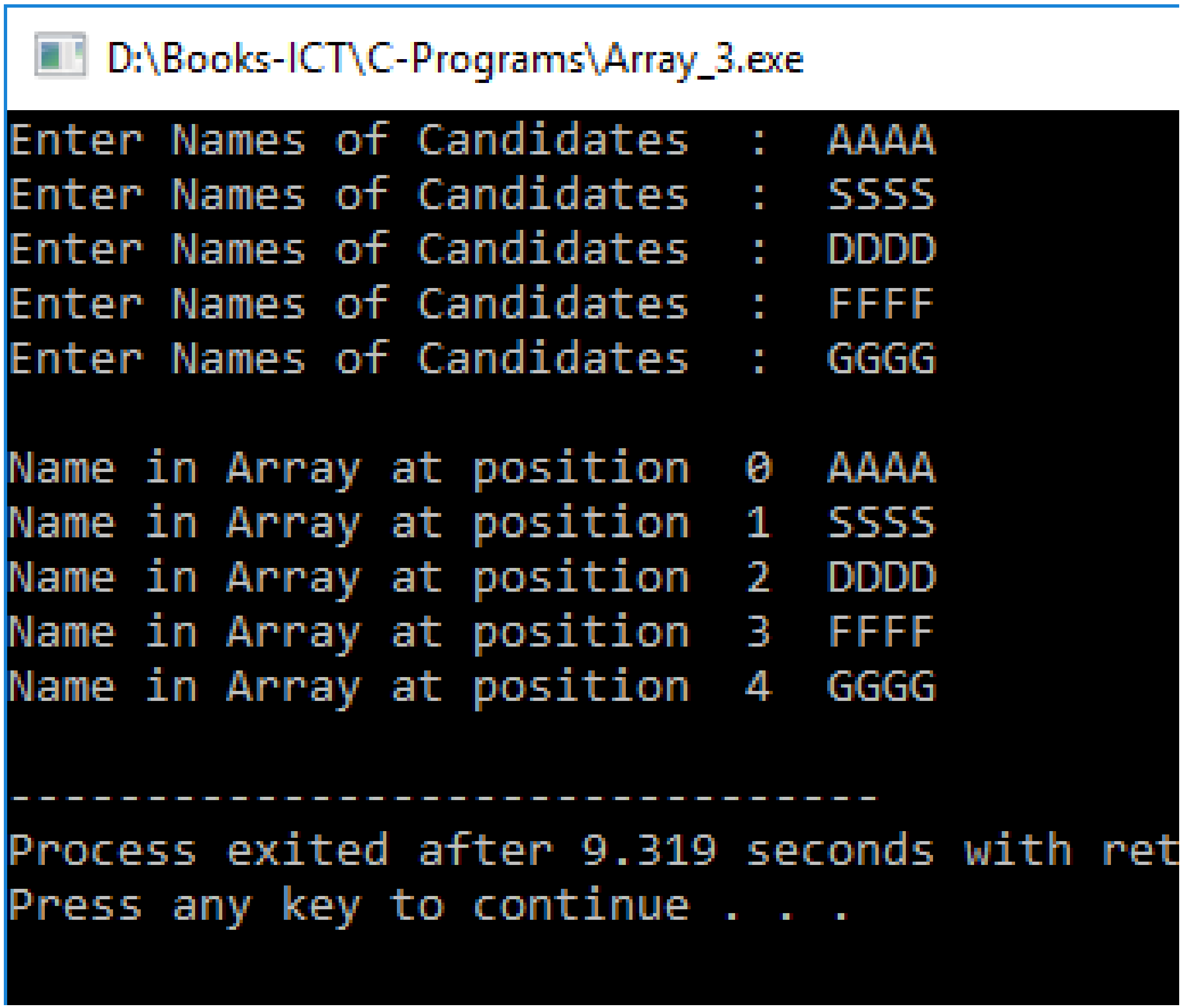
PROGRAMMING FUNDAMENTALS

Spring 2024

21st March

Write a program for a string array



Points to note

- If you leave an element uninitialized, you must leave all the elements that follow it uninitialized
- C++ does not provide a way to skip elements in the initialization list. For example, the following is *not* legal:

```
int numbers[6] = {2, 4, , 8, , 12}; // NOT Legal!
```

Points to note

- You *must* provide an initialization list if you leave out an array's size declaration
- Otherwise, C++ doesn't know how large to make the array
- An accumulator variable must be set to 0 before it is used to keep a running total or the sum will not be correct

Ex # 1

- Initialize a string array with names of rainbow colors.
- Use switch case structure that accepts an integer value and your code displays the rainbow color based on the integer value entered by the user
- Sequence of colors: VIBGYOR
- If user enters 4 → Green will be displayed

Ex # 2

- Define two integer arrays of size ten. The first array has first 10 even numbers. The second integer array has first 10 odd numbers

- The third array has sum of values from first 2 arrays
- Display three arrays in the following format:

Even	Odd	Sum
2	1	3
4	3	7
6	5	11

So on…

Ex # 3

- Write a C++ program that declares an array of **20** elements of type **double**. Initialize the array so that the first 10 elements are equal to the square of the index variable, and the last 10 elements are equal to cube of the index variable. Output the array so that 10 elements per line are printed.

Solution Ex # 3

```
#include <iostream>
using namespace std;
int main()
{
int my_array[20];
int i;
        for (i = 0; i <10; i++)
        {
                my_array[i]= i*i;
                }
        for (i = 10; i <20; i++)
        {
                my_array[i]= i*i*i;
        }
        for (i = 0; i <20; i++)
        {
                cout<<my_array[i]<<" ";
                if ((i+1)%10 == 0)
                cout<<endl;
        }
  return 0;
}
```

Ex#4 To find the smallest value in array

```
int count;
int lowest;
lowest = numbers[0];
for (i = 1; i < SIZE; i++)
{
   if (numbers[i] < lowest)
   lowest = numbers[i];
}
```

Ex # 5

- Write program to search a number in an array
- User enters an input and your code searches in the array.
- Output: value found at position

  or

  value NOT found in array

Array_Search.cpp

Random number function

- The function, rand will return a random number
- The number can be between 0 and 32767
- We can use this function as:

```
x = rand ( );
```

- The random function generates an integer which is assigned to variable x

Random numbers

- Suppose we have a die with six faces marked with 1, 2, 3, 4, 5 and 6
- We want to generate a number between 1 and 6 inclusive

```
rand ( ) % 6;
```

- When 6 divides any number, the remainder will always be less than 6. Therefore, the result will be between 0 and 5 inclusive
- We want the number between 1 and 6, we will add 1

```
1 + rand ( ) % 6;
```

Ex # 6

- Fill the array of size 10 with random numbers generated between 1 and 50 inclusive
- Ask the user to enter a positive integer
- Search in the array
- Display the message accordingly and the complete array

Random_Array.cpp

Ex # 7

- The sales division of a company generates 58 percent of total sales. Write a program that will calculate how much the division will generate if the company has Rs.86000 in sales this year.

Ex # 8

- Write a program that will compute the total sales tax on a purchase. Assume the tax rate is 16%. Your program allows the user to enter sales amount until a sentinel value is entered. It will calculate the tax amount and display the final bill for each purchase transaction.

Lets practice more..

Write a program that allows the user to enter the last names of five candidates in a local election and the number of votes received by each candidate. The program should then output each candidate's name, the number of votes received, and the percentage of the total votes received by the candidate. Your program should also output the winner of the election. A sample output is:

```
Candidate          Votes Received       % of Total Votes

Johnson                 5000                  25.91
Miller                  4000                  20.73
Duffy                   6000                  31.09
Robinson                2500                  12.95
Ashtony                 1800                   9.33
Total                   19300

The Winner of the Election is Duffy.
```

References

- Tony Gaddis
- Lecture → chp 4